structure can still be substantially warped—sometimes across large fractions of layers. This discrepancy reflects the fact that centroidal and regression-based measures capture global alignment, whereas stress and distortion expose finer-grained deviations in how relative distances between emotions are preserved. Thus, high apparent alignment at the aggregate level can coexist with local irregularities in the geometry of emotional spaces; however, linear probing shows that these spaces can still usefully and predictably predict emotion, even if there is some distortion between the synthetic and human-written spaces. Taken together, these results indicate that the claims concern the directional organization of the emotional subspace rather than strict isometry, and that the local warping revealed by stress and distortion is expected in high-dimensional compression while remaining compatible with a functionally coherent and manipulable emotional manifold.

4.3 Model Psychology

Having established the external consistency of emotional geometry across datasets, we now turn inward, asking how these emotions are internally structured within the model, and what this reveals about the model's implicit psychological architecture.

The first perspective examined is neural encoding patterns using ML-AURA (Section 3). Results focus on Llama3.1-8B-Base, with replications across models in Appendix F. Across the six Ekman emotions, an average of 75% of neurons per layer exceed the selectivity threshold, with sadness (98%) and surprise (97%) most pervasive, and fear lower (48%). This reflects sparse specialization rather than weak separability, and, importantly, selectivity rather than activation magnitude: ML-AURA identifies neurons whose responses discriminate emotions, not neurons that merely fire strongly. Non-Ekman emotions—envy, neutral, excitement—also show strong separability, averaging 88%.

When analyzing by architectural component, MLP layers show slightly higher selectivity than attention layers (79% vs. 76.5%). Selectivity fluctuates across depth, with no clear monotonic trend: while the first layer starts at 76% and the final layer ends at 76.3%, several peaks and troughs occur in between, with the highest selectivity observed at layer 26 (79%). These patterns support the conclusion that emotional information is not confined to late layers or specialized regions, but is instead distributed broadly and redundantly throughout the network. These patterns are visualized by emotion in a layer-by-layer fashion in Figure 1.

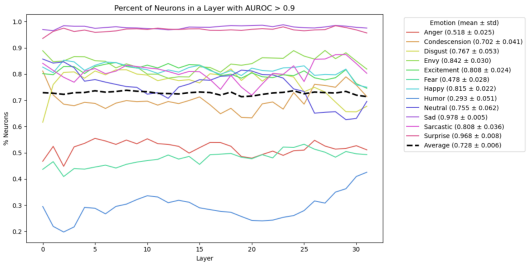

Figure 1: Results of ML-AURA by layer and emotion. Results are in terms of percent of neurons with an AUROC score above 0.9.

To understand how emotions are geometrically represented in the network, we examine their structure within the derived SVD subspace. This subspace provides a low-dimensional lens into the model's internal affective organization. Our first goal is to assess how consistently emotions are arranged along the principal axes across layers and layer types.

We find that the emotional structure is remarkably stable across the Llama3.1-8B-Base model, particularly for the top three components. Across layers, the average Spearman correlation in emotion rankings is 0.87, 0.83, and 0.80 for PC1, PC2, and PC3, respectively; the corresponding Kendall's Tau values are 0.82, 0.77, and 0.74. These results indicate that, while the magnitude and orientation of the components may shift, their semantic content remains intact.

Even when using a more fine-grained labeling scheme, as in the Go-Emotions dataset, which contains nearly three times as many emotion categories, we observe similar consistency. Rank-

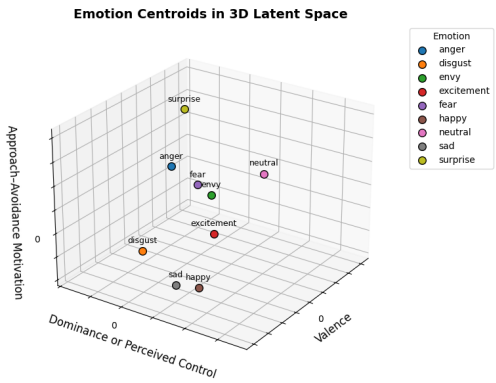

Figure 2: Emotion centroids plotted on the emotional axis found

order correlations for Go-Emotions along the top three PCs remain high: Spearman values of 0.92, 0.74, and 0.73, and Kendall's Tau of 0.86, 0.68, and 0.68. These findings reinforce the conclusion that the model's emotional manifold is structurally stable, with interpretable axes. Appendix G reproduces the high consistency in how emotions are arranged along principal axes across all models studied.

Having established the stability of the SVD subspace across layers and datasets, the semantic content of the leading principal components is examined. Figure 2 visualizes the first three emotion axes that we describe below.

- PC1 strongly resembles a valence dimension. Emotions such as happy, surprise, and excitement lie at the positive end, while anger and fear occupy the negative end—suggesting a pleasure–displeasure continuum common to many psychological models.
- PC2 appears to reflect dominance or perceived control. Emotions high on this axis (e.g., fear, sadness) are often associated with low control or submission, whereas those at the opposite end (e.g., happy, surprise) may reflect more autonomous or socially detached states.
- PC3 maps onto approach–avoidance motivation. Emotions like excitement, happy, and envy—typically associated with goal-seeking behavior—score high, while anger and fear, more linked to avoidance or defensive responses, score low.
- PC4 may correspond to arousal or urgency. Surprise and fear rank highly, consistent with high physiological activation, while happy and neutral lie at the calmer end.

These dimensions are not explicitly supervised, but show surface-level resemblance to constructs proposed in affective science, such as valence, arousal, dominance, and approach–avoidance tendencies (cf. Russell (1980); Mehrabian (1996); Davidson (1995)). While these alignments are not exact, and many components blend multiple emotional signals, the emergence of such patterns suggests that large language models may implicitly encode affective distinctions that overlap with long-standing psychological taxonomies. This correspondence invites further investigation into the extent to which models trained solely on text internalize latent emotion structures, and whether these can serve as proxies or tools for understanding affective semantics in language.

Figure 3 provides a t-SNE visualization of the emotional space. Despite the dimensionality reduction, emotion classes form distinguishable, partially overlapping clusters, with closely related emotions (e.g., happy and excitement) frequently co-localized and others (e.g., fear and joy) appearing more spatially distant. While not all boundaries are sharp, the observed structure reinforces earlier findings: emotional information is embedded in a distributed yet semantically coherent geometry.

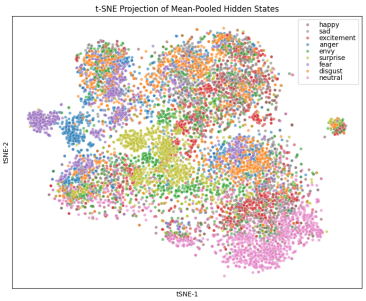

Figure 3: Plot of projected emotions in t-SNE space.

Together, the distributed AUROC patterns, stable subspace directions, interpretable principal components, and emergent clustering structure suggest that LLMs encode emotion not as isolated tags, but as coherent, multidimensional structures—akin to a learned affective manifold.

4.4 Steering Emotion Representations

Prior work on emotional steering in LLMs focuses primarily on shifting the emotional tone of generated text. (Subramani et al., 2022) and (Zou et al., 2023) learn vectors to modify output valence or categorical emotion. More recently, (Hollinsworth et al., 2024) attempts to steer internal emotional representations but collapses emotion into a binary positive/negative axis, achieving valence shifts 53.5% of the time. In contrast, we aim for fine-grained control over the model's internal perception of emotion across a full categorical space, while preserving semantic content.

We train a module that operates within the previously constructed SVD-based emotional subspace. For each emotion, we select all layers and sublayers where adding the centroid direction to same-emotion hidden states improves 1-vs-all classification AUROC beyond a fixed threshold. These layers

Model	Language	Sad	Happy	Fear	Anger	Neutral	Disgust	Envy	Excitement	Surprise	Overall
Llama3.1-8B	English	22 → 99 (0.14)	20 → 92 (0.23)	7 → 68 (0.25)	21 → 58 (0.21)	1 → 96 (0.21)	0 → 88 (0.24)	0 → 92 (0.24)	0 → 90 (0.23)	8 → 64 (0.22)	9 → 83 (0.22)
Llama3.1-8B	Non-English	9 → 100 (0.12)	13 → 88 (0.21)	5 → 68 (0.21)	14 → 65 (0.20)	16 → 95 (0.20)	3 → 94 (0.21)	0 → 99 (0.21)	1 → 88 (0.21)	7 → 64 (0.19)	**8 → 84 (0.20)**
Olmov2	English	27 → 93 (0.09)	32 → 98 (0.07)	2 → 75 (0.05)	21 → 100 (0.07)	6 → 100 (0.04)	1 → 99 (0.04)	0 → 91 (0.04)	0 → 81 (0.07)	7 → 83 (0.07)	11 → 91 (0.06)
Olmov2	Non-English	33 → 84 (0.08)	21 → 80 (0.07)	2 → 59 (0.04)	23 → 100 (0.06)	8 → 99 (0.05)	1 → 98 (0.04)	0 → 84 (0.04)	4 → 73 (0.08)	4 → 91 (0.07)	11 → 86 (0.06)
Ministral	English	17 → 76 (0.09)	24 → 90 (0.10)	5 → 98 (0.09)	17 → 100 (0.10)	16 → 100 (0.10)	2 → 96 (0.10)	0 → 99 (0.10)	1 → 82 (0.10)	7 → 99 (0.10)	10 → 91 (0.10)
Ministral	Non-English	14 → 47 (0.09)	28 → 60 (0.09)	5 → 90 (0.08)	22 → 97 (0.09)	16 → 95 (0.09)	3 → 87 (0.10)	0 → 90 (0.10)	3 → 51 (0.09)	4 → 88 (0.09)	**11 → 79 (0.09)**

Table 2: Top-1 prediction rates before and after learned steering for each target emotion across datasets and the semantic similarity loss between the hidden-state representations before and after steering. Each cell shows *baseline → post-steering (average semantic similarity loss)* accuracy.

are used for steering and serve as a proxy for the more challenging task of controllable emotional manipulation. At each selected layer, the model's hidden state is projected into the emotional latent space. The projected representation is passed through a one-layer MLP with a GELU activation to compute a shift, which is then mapped back into hidden-state space and added residually. The MLP is trained to steer the model's representation to favor the target emotion token when prompted.

We define the overall training objective as: $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{token}} + \mathcal{L}_{\text{sem}}$ where $\mathcal{L}_{\text{sem}}$ preserves semantic meaning and $\mathcal{L}_{\text{token}}$ enforces perceptual control.

Semantic Preservation. The semantic consistency loss combines cosine and ℓ_2 distance between the original and shifted final-layer hidden states:

$$\mathcal{L}_{\text{sem}} = (1 - \cos(h_{\text{base}}, h_{\text{shifted}})) + \gamma \cdot \frac{\|h_{\text{base}} - h_{\text{shifted}}\|_2}{\|h_{\text{base}}\|_2 + \|h_{\text{shifted}}\|_2}$$

Emotion Control. To ensure accurate emotion classification, we combine a standard cross-entropy loss with a token-level margin loss. The margin loss enforces that the logit for the target emotion token e_i exceeds its synonyms s_i by a margin m_1 (0.5), and that both exceed all other emotions e_j and their synonyms by m_2 (10):

$$\mathcal{L}_{\text{margin}} = \max(0, m_1 - (\log p_{\text{e}_i} - \log p_{\text{s}_i})) + \max(0, m_2 - (\log p_{\text{s}_i} - \log p_{\text{e}_j}))$$

To prevent the model from optimizing by suppressing unrelated tokens, we weight the loss for emotion tokens more heavily in $\mathcal{L}_{\text{CE}}$: $\mathcal{L}_{\text{token}} = \mathcal{L}_{\text{CE}} + \lambda \cdot \mathcal{L}_{\text{margin}}$

We optimize the objective using AdamW with learning rate 1e-3 and weight decay 1e-2, using a cosine schedule with 50 warm-up steps. Steering uses the top 40 dimensions of the centered SVD-derived emotional subspace. The learned module is trained independently for each target emotion across all selected steering layers, using supervision from emotion-token prompts and hidden-state consistency targets. At evaluation, token sampling is performed with temperature 0 for determinism.

The steering module is evaluated by measuring how reliably it shifts the model's emotion classification toward the target label while preserving semantic similarity. Table 2 shows that the learned steering approach achieves consistent and accurate control of internal emotional representations across English and non-English datasets for all three models. For most emotions, post-steering accuracy exceeds 80%, with many cases reaching 90–100%. Even for harder emotions and lower-resource languages, steering reliably improves target classification, with nearly all settings achieving at least 50% accuracy. Semantic similarity losses remain low across models and languages, indicating that steering preserves the core representational content while shifting affective interpretation. LLaMA-3.1-8B is the most steerable model across a plurality of settings, while OLMo-v2 shows the largest average improvement—despite OLMo-v2-Instruct previously exhibiting substantial geometric misalignment in Section 4. Appendix H reports full results per dataset, and Appendix I provides ablations isolating key methodological factors. Table 3 illustrates how emotional steering manifests in practice, with inputs rewritten to express the target emotion.

5 Conclusion

Using a combination of probing, alignment, and causal intervention techniques, this work shows that emotional representations in LLMs are directionally consistent across layers, datasets, and languages. We find that emotions cluster in coherent, low-dimensional subspaces whose structure is stable across architectural depth and transferable across linguistic and cultural domains. The leading axes of this